

Machine Learning Results and Interpretation

Introduction

In this stage, the objective is not only to apply machine learning models, but also to critically evaluate whether the statistical and structural relationships identified during the exploratory data analysis can be transformed into meaningful predictive signals.

The problem is formulated as a binary classification task, where the goal is to predict the direction of the next-day portfolio return as either positive or negative.

This formulation reflects a realistic financial decision-making scenario because investors are often more interested in predicting the direction of movement rather than the exact magnitude of return.

Dataset Preparation

The dataset is divided into:

- Training set: 1239 observations
- Test set: 310 observations

A time-based split is used instead of a random split.

This is important because financial data follows a chronological order. The model should be trained on past data and tested on future data.

Using a time-based split avoids look-ahead bias and makes the evaluation more realistic.

Feature Selection

The feature set includes:

- BIST100 returns
- Lagged portfolio returns
- Rolling volatility
- Short-term rolling trend indicators
- Market-related variables

These features are not selected randomly. They are directly motivated by the findings of the EDA stage.

During EDA, the analysis showed:

- Strong market dependence
- Volatility clustering
- Heavy-tailed return distributions

- Time-varying risk behavior

Therefore, the selected features aim to capture both market-wide effects and recent historical behavior.

Machine Learning Models

Three machine learning models are implemented to evaluate different levels of complexity.

Logistic Regression

Logistic Regression is used as the linear baseline model.

It assumes that the relationship between input variables and the probability of a positive return can be represented through a linear decision boundary.

This model is useful because it provides a simple benchmark for evaluating whether the prediction problem has a clear linear structure.

Random Forest

Random Forest is used as a non-linear ensemble model.

It combines multiple decision trees and can capture:

- Non-linear relationships
- Feature interactions
- Complex market behavior

This model is expected to perform better than Logistic Regression if the relationship between financial indicators and return direction is non-linear.

Gradient Boosting

Gradient Boosting is included as a more advanced boosting-based model.

It builds trees sequentially and attempts to correct the errors of previous models.

This model is generally powerful, but it may also be sensitive to noise, especially in financial datasets where the signal-to-noise ratio is low.

Model Performance Results

The results show that all models struggle to produce strong predictive performance.

This is an important finding because financial market data is noisy, dynamic, and difficult to predict using historical information alone.

Logistic Regression Results

Logistic Regression achieved:

- Accuracy: approximately 0.50
- ROC-AUC: approximately 0.494

This indicates that the model performs almost at the level of random guessing.

The result suggests that the relationship between the selected input features and next-day return direction is not linearly separable.

In other words, there is no simple linear boundary that can reliably distinguish between positive and negative return days.

This finding is consistent with the EDA results, where return distributions were found to be:

- Non-normal
- Heavy-tailed
- Highly volatile
- Affected by extreme values

Random Forest Results

Random Forest achieved:

- Accuracy: approximately 0.516
- ROC-AUC: approximately 0.504

This model produced the best performance among the three models, but the improvement over Logistic Regression was very small.

The ROC-AUC value being close to 0.5 indicates that the model has almost no strong discriminative power.

This means that even when non-linear relationships and feature interactions are considered, the underlying predictive signal remains weak.

From a financial perspective, this suggests that past market information contains limited predictive content for future return direction.

Gradient Boosting Results

Gradient Boosting achieved:

- Accuracy: approximately 0.464

- ROC-AUC: approximately 0.472

This model performed worse than both Logistic Regression and Random Forest.

Although Gradient Boosting is generally considered a strong predictive model, its weaker performance in this project suggests that it may be overfitting noise.

Financial return data often contains a low signal-to-noise ratio, meaning that many apparent patterns may not persist in future observations.

Therefore, a more complex model does not necessarily produce better results.

Classification Report Interpretation

The classification reports provide additional insight into model behavior.

For the Random Forest model, recall for the positive class is slightly higher than recall for the negative class.

This means that the model is somewhat more sensitive to detecting positive return days.

However, precision remains moderate, meaning that many predicted positive signals are incorrect.

This creates both:

- False positives
- False negatives

As a result, the model does not consistently identify true market movements.

Confusion Matrix Interpretation

The confusion matrix illustrates the limitations of model prediction.

The model correctly identifies some positive and negative observations, but a large number of misclassifications remain.

Specifically, the model:

- Incorrectly labels many negative returns as positive
- Fails to capture a significant portion of actual positive returns

This balanced but weak structure indicates that the model captures some limited tendencies in the data, but does not have strong predictive power.

Connection Between EDA and ML Results

A key insight emerges when the machine learning results are compared with the EDA findings.

The EDA stage showed:

- Strong correlation between stocks and BIST100
- Significant beta exposure to market movements
- Strong co-movement among Turkish large-cap stocks

However, the machine learning models showed weak predictive performance.

This means that the observed relationships are mostly contemporaneous rather than predictive.

In other words:

- Stocks may move together at the same time
- But this does not mean today's information can reliably predict tomorrow's return direction

This explains why machine learning models struggle despite strong correlations observed during EDA.

Effect of Heavy-Tailed Returns

The heavy-tailed nature of returns plays an important role in limiting model performance.

Earlier statistical tests, such as the Jarque-Bera test, confirmed that return distributions are not normal.

Extreme returns have a significant impact on financial outcomes, but they are rare and difficult to predict.

Machine learning models tend to focus on average patterns and therefore often fail to capture extreme events effectively.

This reduces prediction reliability, especially during volatile periods.

Rolling Prediction Error Analysis

Rolling prediction error analysis shows that model performance is not stable over time.

During relatively calm market periods:

- Prediction errors are smaller
- Models perform slightly better

During high-volatility periods:

- Prediction errors increase
- Model reliability decreases

This result aligns with the volatility clustering observed in the EDA stage.

It also highlights the dynamic nature of financial markets, where predictive relationships change over time.

Feature Importance Interpretation

Feature importance results from the Random Forest model reinforce the findings from the EDA stage.

BIST100-related variables emerge as the most influential predictors.

This confirms that local market movements dominate stock behavior.

Lagged returns and rolling volatility measures also contribute to the model, indicating that recent market conditions contain some limited predictive information.

However, USD/TRY and S&P500 variables have relatively lower importance.

This suggests that macroeconomic and global factors play a secondary role in short-term prediction compared to domestic market conditions.

Overall Machine Learning Interpretation

The machine learning analysis shows that models are able to identify weak patterns in the dataset.

However, these patterns are not strong enough to generate reliable predictions.

The low accuracy and ROC-AUC values indicate that a large portion of return variability is driven by factors not captured in the dataset, such as:

- Unexpected news
- Investor psychology
- External shocks
- Political uncertainty
- Sudden macroeconomic changes

Therefore, the machine learning models are more useful for understanding the structure and limitations of financial prediction than for producing highly accurate short-term forecasts.

Final Interpretation

The machine learning analysis confirms that stock returns are primarily influenced by local market conditions, especially BIST100-related variables.

However, these relationships do not translate into meaningful predictive power for next-day returns.

All models perform close to random classification, indicating that short-term market movements are largely unpredictable using historical market data alone.

These results support the idea that financial markets are highly efficient in the short term.

Prices rapidly incorporate available information, leaving limited opportunity for systematic prediction based only on past data.

Conclusion of Machine Learning Stage

The findings of this stage are consistent with the exploratory data analysis and statistical modeling results.

The project shows that:

- Turkish stocks are strongly dependent on BIST100 market behavior
- Return distributions are heavy-tailed and volatile
- Volatility changes over time
- Strong correlation does not guarantee predictability
- Machine learning models provide limited short-term forecasting power

Overall, the machine learning stage creates a coherent narrative with the EDA results.

The combination of:

- Strong market dependence
- Heavy-tailed distributions
- Volatility clustering
- Low predictive accuracy

highlights the complex and uncertain nature of financial markets.